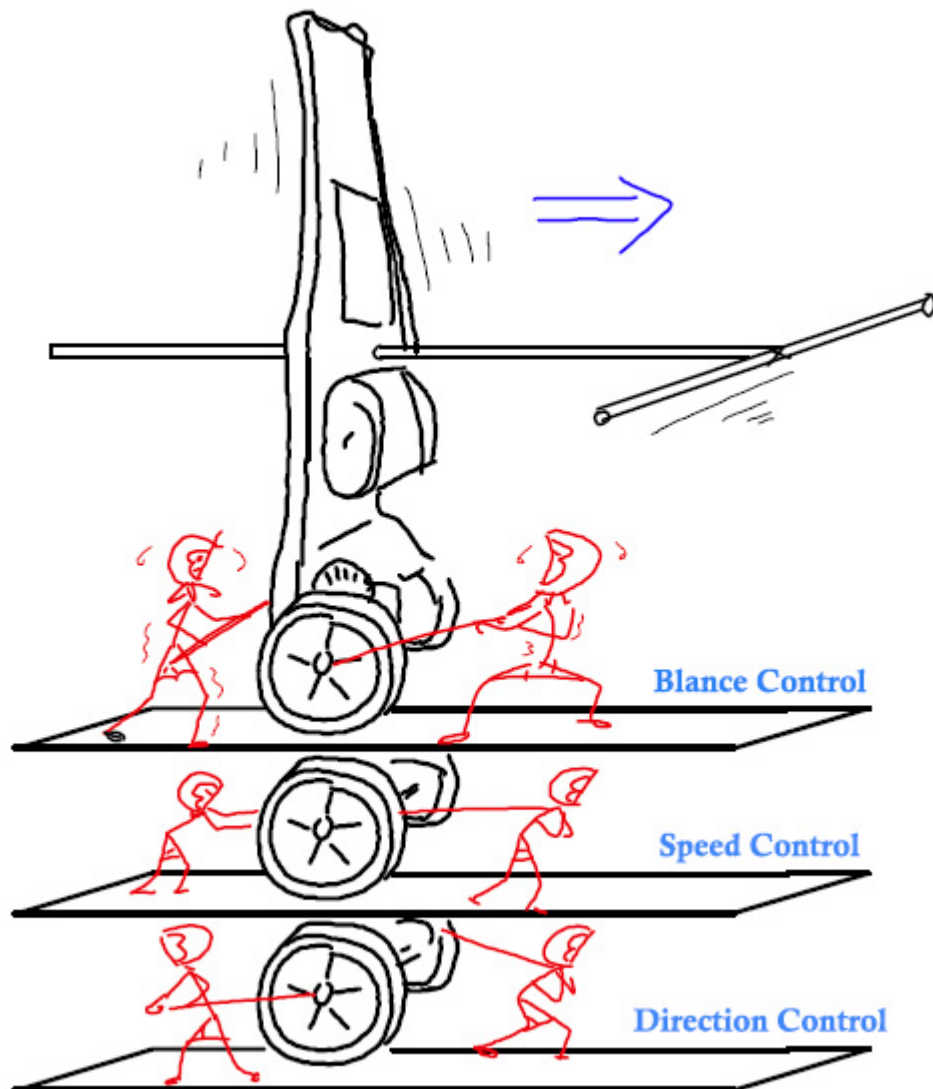


Lesson 1 Tumbler control principle



Tips: If you have any questions or run into any problems during assembling and testing Tumbler please feel free to contact us at service@elegoo.com or euservice@elegoo.com (Europe customers).

I . Analysis of Tumbler on motion control.

There are 3 basic controls for Tumbler motion control task.

1. Balance Control:

To keep the Tumbler maintaining the upright balance by controlling its forward and reverse motion

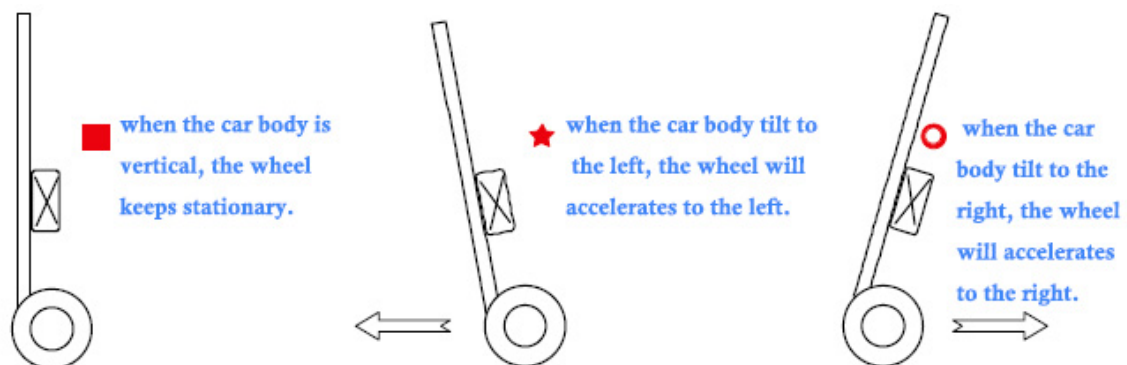
2. Speed Control:

To control the speed by adjusting the inclination of the Tumbler, and in fact it finally evolves to control the wheel speed by controlling the rotation speed of the motor.

3. Direction Control:

To control the direction change of the Tumbler by controlling the rotational differential between the two motors.

II . Balance Control of Tumbler



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When the car body is vertical to the wheel, keep still

When the car body tilt to the left, the wheel will accelerates to the left. Once the speed of the car wheel is greater than car body, the car body will return to the vertical state to the wheel.

When the car body tilt to the right, the wheel will accelerates to the right. Once the speed of the car wheel is greater than car body, the car body will return to the vertical state to the wheel.

Formula of the Wheel Acceleration Control:

$$a = k1 * \text{angle} + k2 * \text{angular_velocity}$$

In the formula, a is the acceleration control of Tumbler, angle is the inclination of Tumbler, angular velocity is angular velocity of Tumbler, K1 and K2 is the proportion coefficient.

Among which, K1 determines whether the car can return to the vertical balance position, and it must greater than gravity acceleration; K2 determines the time that Tumbler will take to return to vertical balance position. Appropriate k2 can make Tumbler to stabilize in a vertical position as quick as possible.

In a word,

1. The bigger the tilt angle is, the quicker the Tumbler will tilt, and the bigger acceleration the wheel need to keep the car balance.
2. The direction of the movement of the Tumbler is the same as the direction of the tilt

III . The measurement of the tilt and angular velocity

We can get a conclusion from the formula if we want to control the balance of Tumbler, we

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need to measure the value of tilt and angular velocity.

The MPU6050 sensor is a chip that integrates a 3-axis gyroscope and a 3-axis accelerometer. It can communicate with Arduino via bus.

To measure the angular velocity by gyro, and to measure the angular acceleration by accelerometer, then after the Kalman filter algorithm based on original 6-axis measurement, we got the accurate and stable balance value of tilt and angular velocity of Tumbler.

IV . Speed Control of Tumbler

The speed of the Tumbler can be obtained by encoder measurement.

IV . Direction Control of Tumbler

The rotation control amount of Tumbler can be obtained by measuring the angle of Z-axis.

Core code preview for balance control:

```
void balanceCar()
{
    sei();
    encoder_left_pulse_num_speed += pwm_left < 0 ? -encoder_count_left_a : encoder_count_left_a;
    encoder_right_pulse_num_speed += pwm_right < 0 ? -encoder_count_right_a :
encoder_count_right_a;
    encoder_count_left_a = 0;
    encoder_count_right_a = 0;
    mpu.getMotion6(&ax, &ay, &az, &gx, &gy, &gz);
    kalmanfilter.Angle(ax, ay, az, gx, gy, gz, dt, Q_angle, Q_gyro, R_angle, C_0, K1);
    kalmanfilter_angle = kalmanfilter.angle;
    double balance_control_output = kp_balance * (kalmanfilter_angle - angle_zero) + kd_balance *
(kalmanfilter.Gyro_x - angular_velocity_zero);
```

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```

speed_control_period_count++;
if (speed_control_period_count >= 8)
{
    speed_control_period_count = 0;
    double car_speed = (encoder_left_pulse_num_speed + encoder_right_pulse_num_speed) * 0.5;
    encoder_left_pulse_num_speed = 0;
    encoder_right_pulse_num_speed = 0;
    speed_filter = speed_filter_old * 0.7 + car_speed * 0.3;
    speed_filter_old = speed_filter;
    car_speed_integral += speed_filter;
    car_speed_integral += -setting_car_speed;
    car_speed_integral = constrain(car_speed_integral, -3000, 3000);
    speed_control_output = -kp_speed * speed_filter - ki_speed * car_speed_integral;
    rotation_control_output = setting_turn_speed + kd_turn * kalmanfilter.Gyro_z;
}

pwm_left = balance_control_output - speed_control_output - rotation_control_output;
pwm_right = balance_control_output - speed_control_output + rotation_control_output;

pwm_left = constrain(pwm_left, -255, 255);
pwm_right = constrain(pwm_right, -255, 255);

if (motion_mode != START && motion_mode != STOP && (kalmanfilter_angle <
balance_angle_min || balance_angle_max < kalmanfilter_angle))
{
    motion_mode = STOP;
    rgb.brightRedColor();
    carStop();
}

if (motion_mode == STOP)
{
    car_speed_integral = 0;
    setting_car_speed = 0;
    pwm_left = 0;
    pwm_right = 0;
    carStop();
}
else
{
    if (pwm_left < 0)
    {
        digitalWrite(AIN1, 1);
    }
}

```

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```
    digitalWrite(AIN2, 0);
    analogWrite(PWMA_LEFT, -pwm_left);
}
else
{
    digitalWrite(AIN1, 0);
    digitalWrite(AIN2, 1);
    analogWrite(PWMA_LEFT, pwm_left);
}
if (pwm_right < 0)
{
    digitalWrite(BIN1, 1);
    digitalWrite(BIN2, 0);
    analogWrite(PWMB_RIGHT, -pwm_right);
}
else
{
    digitalWrite(BIN1, 0);
    digitalWrite(BIN2, 1);
    analogWrite(PWMB_RIGHT, pwm_right);
}
}
}
```

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